

From Reverse Entailment Diagnostic to the Universal Linguistic Domain

A slot-geometric architecture for nominal potential, exeme placement, and cross-domain modeling

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Abstract

This paper provides a rigorous theoretical “lift” from **Reverse Entailment Diagnostic (RED)**—a bidirectional method for testing **nominal potential**—to a background architectural (“geometrical”) hypothesis: the **Universal Linguistic Domain (ULD)**. RED operationally tests whether linguistic material can be **packaged** into a noun-entity and verified in **noun-slot** positions (e.g., subject/object), and it treats successful packaging as evidence that the base has latent nominal capacity. My RED manuscript (“A Universal Noun Slot in the Linguistic Mind: Reverse Entailment Across All Parts of Speech with Computational Applications”) explicitly defines this as a 3-step procedure: nominal derivation → interpretability → noun-slot occupancy. The central claim of the present article is an inference-to-the-best-explanation: if RED yields stable,

cross-category and cross-linguistic results (via possessive/genitive and nominalizing resources), then RED presupposes a stable target **position-type** (a universal noun slot) and, more generally, a structured (“geometrical”) space of position-types and licensed transitions. ULD, Universal Linguistic Domain, names that space. The paper (i) derives a minimal set of ULD slot families (referential, predication, operator, linking, packaging), (ii) integrates **exeme** as a modality-neutral expression substrate that becomes lexeme-like **by placement** into ULD regions rather than by prior “association,” and (iii) formalizes ULD as a typed graph in which RED corresponds to the existence and cost of specific paths (notably PACK → REF). The analysis is supported by a submission-ready multilingual example set based on my RED dataset (English, Mandarin, Tongan, Spanish, Arabic), and it concludes with computational and neurocognitive predictions linking slot transitions to parsing/SRL improvements and to measurable integration/unification signatures.

Keywords: nominal potential; nominalization; diagnostics; universal noun slot; Universal Linguistic Domain; exeme; slot families; constructional typing; computational linguistics; neurolinguistics

1. Introduction

A persistent empirical fact in linguistic theory is that materials that are not canonical nouns— verbs, adjectives, clauses, even function words— can nevertheless be construed as **entity-like** in appropriate environments. My RED manuscript makes this claim maximally strong: *all parts of speech* can be nominalized into syntactical noun entities occupying noun-slot positions, with cross-linguistic support from possessive/genitive resources (English -’s, Mandarin **de**, Tongan ’a, Spanish **de**, Arabic **-i**).

RED is designed to transform this observation into a diagnostic: a derived form (e.g., possessive or nominalized) is used to **reverse-entail** the latent nominal potential of the base, and then the base (or minimally packaged base) is tested in noun-slot positions (subject/object).

This paper argues that a diagnostic with these success conditions cannot remain “standalone.” If RED is a stable probe, it presupposes a stable *host architecture*— a structured (“geometrical”) space of position-types and transitions in which “noun-slot occupancy” is meaningful. My ULD manuscript states this in the intended direction: “ULD is the minimal structural hypothesis required to interpret RED.”

1.1 Novelty and contributions

The contribution is not the observation that nominalization exists (classical work already centers nominalization as theoretically diagnostic; see Chomsky 1970). ([WALS Online](#)) The novelty is the **method-to-architecture inference**:

1. **RED as measurement logic (bidirectional diagnostic)**: nominal packaging is treated as evidence about the *base’s capacity*, not merely as a recategorization of outputs.
2. **RED → universal position-type → ULD**: repeated RED success implies a universal noun slot as a position-type, and a position-type implies a wider **domain of slots + licensed operations**, i.e., ULD.
3. **Exeme integration with corrected directionality**: an **exeme** is a modality-neutral form pattern with no commitment to word status, category, or meaning, and lexemehood emerges by **placement** into ULD regions (rather than “exeme associating itself with a lexeme”).
4. **Formalization and mappings**: ULD is formalized as a typed graph; RED outputs map onto slot-family regions and transition costs.
5. **Cross-domain extensions**: computational applications in parsing/SRL and neurolinguistic predictions about unification/integration follow from treating slot transitions as real representational events.

1.2 Roadmap

Section 2 situates RED/ULD relative to nominalization and category-flexibility traditions. Section 3 defines exeme, RED, and ULD. Section 4 provides the rigorous derivation from RED to ULD and to the minimal slot families. Section 5 presents a submission-ready example set and conventions. Section 6 gives the ULD analysis of the examples. Section 7 provides a compact formal appendix (ULD graph objects; diagnostics-to-families mapping). Sections 8–9 outline computational and neuroscience implications. Section 10 states limitations and future work consistent with standard linguistics-article expectations.

2. Background: nominalization, category flexibility, and slot commitments

Nominalization has long served as a diagnostic locus for grammatical architecture, including the division of labor between lexical entries and functional structure (Chomsky 1970). ([WALS Online](#)) More recent frameworks differ on where category information is introduced. Distributed Morphology (DM) treats roots as category-neutral and derives category via functional heads (e.g., *n*), providing a natural place for nominal packaging. ([Glottolog](#)) Construction Grammar (CxG) treats constructions as form–meaning pairings that can assign roles independently of lexical categories, emphasizing schematic typing and inheritance. ([University of Chicago Press](#))

RED explicitly leverages these tensions: my manuscript contrasts fixed-category assumptions with DM/CxG-style flexibility, and proposes RED as a methodology that “confirms nominal potential” by reversing from derived nominal packages to the base’s capacity to occupy noun-slot positions.

The point of ULD is not to replace DM or CxG, but to identify what RED presupposes regardless of implementation: if noun-slot occupancy is a stable success condition, then “noun slot” must be a stable position-type, not merely a label on a lexical class.

3. Core constructs

3.1 Exeme

This manuscript defines **exeme** as a modality-neutral expression unit: a specific ordered pattern realized as graphemes and/or speech sounds, without commitment to word status, grammatical category, or meaning. Exeme is introduced precisely to speak about *form-only* material without prematurely importing lexeme/lemma/category disputes.

Working assumption (this paper): Exemes are the empirical inputs to diagnostics; slot placement is the architectural mechanism that yields stable grammatical/lexical commitments.

3.2 RED (Reverse Entailment Diagnostic)

My RED article (“A Universal Noun Slot in the Linguistic Mind: Reverse Entailment Across All Parts of Speech with Computational Applications”) defines RED as a bidirectional inference mechanism intended to demonstrate that elements across parts of speech can be nominalized into syntactical noun entities that occupy noun-slot positions (subject/object). It specifies a three-step procedure: (i) apply nominal derivation (possessive/nominalizer), (ii) test grammaticality/interpretability, and (iii) verify noun-slot occupancy by placing the base into subject/object roles.

RED is therefore not merely “X can be nominalized”; it is a measurement protocol whose target is

nominal potential— a latent capacity evidenced by successful packaging and slot occupancy.

3.3 ULD (Universal Linguistic Domain)

My ULD concept states the key bridging claim explicitly: **ULD is the minimal structural hypothesis required to interpret RED**, because RED empirically demonstrates universal noun-slot access, and ULD names the underlying space of slots and operations in which such access is possible.

In the present paper, ULD is treated as a **geometrical linguistic space**: an abstract domain of (i) position-types (slots/subdomains) and (ii) licensed transitions (typing/placement/packaging operations) that jointly constrain how expression material can become referential, predicational, operator-scoped, argument-linked, or recursively embedded.

4. The RED → ULD derivation

This section provides a rigorous route (inference steps) from RED's diagnostic logic to ULD as an architectural postulate.

4.1 Step 1: RED presupposes a stable target position-type

RED's Step 3 is decisive: it verifies noun-slot occupancy by testing the base (or minimally packaged

base) in subject/object positions. This means RED succeeds only if there exists a stable target **position-type** that can host entity-like packages. My ULD concept formulates this as a methodological observation: RED is a probe of a position-type rather than a mere label for “noun.”

4.2 Step 2: Cross-linguistic invariance implies universality at the level of positions, not markers

My RED dataset emphasizes that surface markers differ across languages but the structural effect is invariant: material is packaged and verified as occupying noun-slot roles (subject/object). This invariance supports the “first ascent” already explicit in my ULD text: repeated RED success licenses the existence of a **universal noun slot** as a latent invariant.

4.3 Step 3: A universal slot implies a universal space of slots plus transitions

A single universal slot cannot be self-contained. My ULD concept states the crucial generalization: for RED to be definable, there must exist (i) a space of positions in which noun-slot is one position-type, (ii) operations that move material into that position (nominal derivation / possessive packaging / constructional typing), and (iii) interface constraints linking abstract content to syntactic form, since reverse entailment is stated to leverage the semantic–syntactic interface.

This is the “second ascent”:

RED → universal noun slot → ULD (a universal space of position-types and licensed operations).

4.4 Step 4: Why ULD must minimally contain slot families (not an undifferentiated set of slots)

If noun-slot access is universal, then the rest of propositional/discourse structure forces additional regions. My ULD concept identifies the minimal families as regions supporting predication, operator scope/framing, argument linking, and packaging/embedding. Concretely:

- **Referential region (REF):** positions supporting discourse addressability and tracking (the universal noun slot is a privileged node/cluster within this region).
- **Predicational region (PRED):** positions supporting property/event assertion (what reference is predicated of).
- **Operator region (OP):** positions imposing scope/framing (determiners, quantifiers, tense/aspect/modality, negation).
- **Linking region (LINK):** positions regulating argument realization/mapping and thematic linking (connecting predicate structure to participant structure).
- **Packaging region (PACK):** positions enabling re-typing and embedding (nominalization, relativization, subordination, coordination).

This is not a taxonomy of “parts of speech.” It is a minimal architectural partition: once you posit stable referential positions, you must also posit stable predicational and scopal machinery, plus argument linking and recursion/packaging, to account for the basic compositional capacities of grammar.

4.5 Step 5: Exeme’s theoretical role in the derivation

My ULD concept clarifies the division of labor: **exemes are points placed into the space;**

RED is the method that discovers/justifies ULD by identifying which diagnostics cluster reliably and therefore correspond to real regions of the domain.

Crucially, my newer framing makes lexemehood emergent: when a RED-supported placement becomes stable and conventionalized, an exeme repeatedly occupies a slot-family and acquires entrenched identity/meaning.

5. Data and conventions

5.1 Example formatting and glossing

Non-English examples are presented with morpheme-by-morpheme glosses and idiomatic translations, following standard practice. Interlinear conventions follow the Leipzig Glossing Rules. ([Max Planck Institute](#))

5.2 Core multilingual RED examples (from my RED dataset)

My RED paper (“A Universal Noun Slot in the Linguistic Mind: Reverse Entailment Across All Parts of Speech with Computational Applications”) provides a flagship case: English *to book carefully* → derived *to book carefully’s* (“To book carefully’s advantage is clear”), followed by noun-slot verification (“To book carefully is my priority”). It then provides cross-linguistic counterparts using Mandarin **de**, Tongan **’a**, Spanish **de**, Arabic **-i**.

Below is a numbered set (with standardized gloss abbreviations).

(1) English (PACK via possessive -'s; REF verification)

- a. *To book carefully's advantage is clear.*
- b. *To book carefully is my priority.*

(2) Mandarin (PACK via de; REF verification)

- a. 小心预订的优点
- b. xiǎoxīn yùdìng de yōudiǎn
- c. careful book DE advantage
- d. 'the advantage of booking carefully'

(3) Tongan (PACK via 'a; REF verification)

- a. ke puke mālie ' a e lelei
- b. COMP book carefully GEN DET good
- c. 'the advantage/good of booking carefully'

(4) Spanish (PACK via de; REF verification)

- a. La ventaja de reservar cuidadosamente es clara.
- b. reservar cuidadosamente es mi prioridad
- c. 'booking carefully is my priority'

(5) Arabic (PACK via -i; REF verification)

- a. miyyizāt ḥajz baḥudhūr-i ...
- b. ḥajz baḥudhūr huwa al-awlawiyya
- c. 'booking carefully is the priority'

6. Analysis: RED results as ULD path evidence

6.1 The core path: PACK → REF

Across (1)–(5), the observed regularity is that a packaging mechanism creates an entity-like unit that can be verified in noun-slot positions (subject/object). In ULD terms, these are witnesses for the existence of a path:

x (base in PRED or otherwise non-REF region) → PACK(x) → REF(x')

RED's "reverse entailment" then treats the existence of a successful package as evidence that the base has latent nominal potential: the base is eligible (under packaging) to enter REF.

6.2 Operator licensing (OP) as region constraints on reference

Entity-like packages typically become fully discourse-addressable when operator resources are available (determiners/quantifiers; case; scope and definiteness). This aligns with standard distinctions between bare predicates and referential NPs. The ULD claim is architectural: OP is a region whose licensing relations constrain what can stably reside in REF.

6.3 Linking (LINK) and participant structure

My RED paper describes noun-slot verification in terms of argument roles (e.g., A1 theme) across languages. This can be interpreted as LINK-region constraints: argument structure and thematic mapping constrain which packages can function as discourse entities in clause-level structures,

consistent with extensive literature on linking and proto-roles (Dowty 1991). (cs.rochester.edu)

6.4 Exeme placement and emergent lexemehood

Under my directionality, exemehood is the form-only substrate, and lexemehood emerges when a particular pattern becomes stably placed in a slot family across contexts. My ULD concept states this explicitly: when RED-supported placement becomes stable and conventionalized, an exeme repeatedly occupies a slot-family and yields entrenched identity/meaning.

7. Formal appendix (compact): ULD as a typed graph; RED as a path diagnostic

7.1 ULD object

Define $ULD = \langle V, E, \tau, \Lambda \rangle$, where:

- V is a set of position-types (slots).
- $E \subseteq V \times V$ are licensed transitions (typing/placement/packaging).
- $\tau: V \rightarrow \{\mathbf{REF}, \mathbf{PRED}, \mathbf{OP}, \mathbf{LINK}, \mathbf{PACK}\}$ assigns slot-family types.
- Λ is a set of constraints (operator requirements, discourse-stability conditions, argument-linking admissibility, etc.).

7.2 RED as a measurement functional

Let **RED**(**x**, **c**) return a triple:

$$\text{RED}(x, c) = \langle \text{fit}, \text{stability}, \text{coercion} \rangle$$

where **fit** is the best-supported slot-family region for **x** under construction/context **c**, **stability** measures whether **x** supports discourse continuation (e.g., anaphora/topic maintenance), and **coercion** measures how much packaging/scaffolding is required.

My ULD concept explicitly proposes this view of RED: it is bidirectional (behavior → region inference; region → diagnostic predictions) and serves as the operational bridge between observable tests and inferred slot families.

7.3 Diagnostics-to-families mapping

- **PACK evidence:** success of nominal derivation / possessive packaging indicates accessibility of PACK and existence of PACK→REF edges.
- **REF evidence:** successful subject/object placement (noun-slot occupancy) indicates REF membership.
- **LINK evidence:** stable thematic realization and argument status across contexts indicates LINK constraints satisfied (cf. proto-role approaches). (cs.rochester.edu)

8. Computational applications

My RED paper explicitly positions RED as useful for NLP— especially syntactic parsing and semantic role labeling— by identifying nominalized phrases as noun-like entities and improving parse updates accordingly.

8.1 ULD-guided feature design

A practical ULD-based system can treat slot-family inference as an intermediate layer:

1. **Parse-level features:** dependency relations, argument positions, nominal modifiers, and packaging markers. Universal Dependencies provides a multilingual standard for such representations. ([ACL Anthology](#))
2. **Role-level features:** SRL signals about participant structure; PropBank provides an established semantic-role scheme for training and evaluation. ([ACL Anthology](#))
3. **Contextual representations:** transformer encoders (e.g., BERT) provide contextual embeddings that can be combined with symbolic RED features. ([ACL Anthology](#))

8.2 Minimal algorithm sketch

Given an input span x and context c :

- Detect candidate packaging triggers (genitives/nominalizers; language-specific lists).
- Propose candidate ULD paths (e.g., PACK→REF; OP attachment; LINK constraints).
- Score $\langle \text{fit, stability, coercion} \rangle$ and output slot-family labels plus a “path witness.”

8.3 Evaluation targets

- Nominal-potential classification (RED-positive vs RED-negative).
 - Slot-family labeling accuracy (REF/PRED/OP/LINK/PACK).
 - Downstream gains in multilingual parsing and SRL on nominalization-heavy constructions (compared with UD/SRL baselines). ([ACL Anthology](#))
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9. Neuroscience and processing predictions

My RED paper (“A Universal Noun Slot in the Linguistic Mind: Reverse Entailment Across All Parts of Speech with Computational Applications”) already sketches neurological grounding: noun phrases and nominalized forms recruit left inferior frontal regions associated with syntactic unification, and nominalized verbs may show convergent activation across object/motor systems. This aligns with broader neurolinguistic frameworks treating LIFG as central to unification (Hagoort 2005) ([PubMed](#)) and with evidence linking action-language processing to distributed motor-related assemblies (Pulvermüller 2005). ([PubMed](#))

9.1 ULD path-length hypothesis (ERP/MEG/fMRI)

If PACK→REF transitions are genuine representational events, then the cost of coercing non-REF material into REF should correlate with integration demands. A straightforward prediction is that marked packaging contexts will show stronger semantic integration signatures (e.g., N400 modulation) than canonical nominal uses, controlling for lexical expectancy (Kutas & Federmeier 2011). ([PMC](#))

9.2 Unification-load hypothesis

Packaging that creates a discourse-addressable referent should increase unification demands relative to baseline predication, predicting increased LIFG engagement with higher coercion values, consistent with unification-centered accounts of LIFG. ([PubMed](#))

10. Limitations and future work

Standard linguistics-article practice expects an explicit limitations/future-work section rather than leaving constraints implicit. The present preprint has four primary limitations:

1. **Data scope:** the multilingual RED set is illustrative and theory-driven; it should be expanded to additional language families and to packaging types beyond genitives/nominalizers (e.g., classifier-driven referentialization).
2. **Formal coverage:** the typed-graph formalization is minimal; future work should define operator constraints and stability metrics with greater precision and connect them to explicit semantic types.
3. **Computational validation:** the algorithmic sketch requires implementation and benchmarking (ablation studies on RED features vs embeddings; multilingual generalization tests using UD and PropBank resources). ([ACL Anthology](#))
4. **Neurocognitive specificity:** predictions are presently directional; they should be tested with controlled stimuli manipulating coercion and path length while holding constant frequency and predictability. ([PMC](#))

11. Conclusion

RED is defined in my work as a 3-step diagnostic— nominal derivation, interpretability, and noun-slot occupancy— that aims to reveal nominal potential as a latent capacity rather than as a surface category label. This paper argued that if RED is stable across categories and languages, it entails a universal noun slot as a position-type, and that a position-type entails a broader architectural space: ULD, the universal domain of slot families and licensed transitions. Integrating exeme yields a coherent micro-to-macro picture: exemes are form-only inputs; RED is the measurement bridge; ULD is the inferred geometry in which stable placement yields emergent lexemehood. The result is a submission-ready framework with explicit novelty (diagnostic-to-geometry inference), a formal backbone (typed graph + RED functional), and clear cross-domain extensions (NLP and neurolinguistics).

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